

- Week 1:** 1. Each of the four cards shown below has a number on one side and a letter on the other side. What is the minimum number of cards that must be turned in order to prove the correctness of this statement: "If a card has a vowel on one side, then it has an even number on the other side?"

A	C	4	7
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Proof. We only need to check card $\boxed{A}$ and $\boxed{7}$. We check $\boxed{A}$ because if that is false, the claim doesn't hold. Also note the contrapositive of *If one side has vowel then there's even number on the other side* will be *If one side has odd number then there's a consonant on the other side*. So we check $\boxed{7}$ as well. The statement says nothing when one side is a consonant so we ignore that. The $\boxed{4}$ card will work either way so also ignored. $\square$

2. Is there a explicit bijection from $\mathbb{N} \rightarrow \mathbb{Q}$ or other way around?

Proof. Yes. Consider f: $\square$

Week 2: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Prove that if f is bounded on every closed interval, then f is bounded on $\mathbb{R}$.

Week 3: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Prove that if f is bounded on every closed interval, then f is bounded on $\mathbb{R}$.